

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 11

Video and audio editing

14 questions from 12 question papers, May/June 2023 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

..... [5]

10 Explain why the following techniques could be needed when making videos.

(a) trimming a video clip

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..... [2]

(b) adding subtitles to a video

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(c) adding sound to a video

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..... [2]

7 Sound editing software has a number of features.

Describe what is meant by:

(a) fade in and fade out

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..... [2]

(b) lossless compression

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(c) lossy compression.

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..... [2]

10 Sound editing software can be used to alter the sampling rate in audio files.

(a) Explain what is meant by the term sampling rate.

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..... [2]

(b) Explain how the sampling rate can affect the audio files.

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..... [2]

8 Sound editing software has a number of features that can be used to edit audio files.

(a) State **two** reasons why an audio clip may need to be trimmed.

Reason 1

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Reason 2

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[2]

(b) State **one** reason why separate audio clips may need to be spliced together.

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[1]

(c) Describe the term equalisation when applied to audio clips.

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[3]

8 Explain why the following techniques could be used when editing a video.

(a) Creating credits

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..... [2]

(b) Extracting a still image

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..... [2]

(c) Exporting a video clip in a different file format

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11 Describe the effects of the different methods of compression on audio quality.

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10 Audio editing software has many features. These can be used when editing an audio clip.

Describe the following terms when applied to audio clips:

(a) normalisation

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..... [2]

(b) noise reduction.

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9626/13 May/June 2025 · Question 6[Mark scheme](#)

- 6 (a)** Give reasons why a person editing a video would use fading effects.

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- (b)** Give reasons why a person editing a video would use captions.

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9626/11 and 9626/13 Oct/Nov 2025 · Question 8[Mark scheme](#)

- 8** Sound files can be edited.

Describe the following features. For each feature, give a reason why it is used.

- (a)** overdubbing

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..... [2]

(b) equalisation

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..... [2]

(c) reverberation

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..... [1]

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..... [4]

9626/11 May/June 2026 · Question 5

Mark scheme

5 A video has been created with an aspect ratio of 16:9.

(a) Describe what is meant by an aspect ratio of 16:9.

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(b) Explain **two** reasons why an aspect ratio of 16:9 rather than an aspect ratio of 4:3 has been chosen.

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9626/11 May/June 2026 · Question 6

Mark scheme

6 A student has to record an audio clip of him delivering a speech and wants to save this file. He can use MP3 or MP4a file formats.

Compare the use of MP3 and MP4a file formats to export this audio clip.

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..... [2]

Mark schemes

Mark scheme · 9626/11 May/June 2023 · Question 6

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Question	Answer	Marks
6(a)	Five from: • Manager checks the space on the target drive/location (1st) – to make sure there is sufficient space for the data (that will be moved) (1) • Manager selects the time they want the back-up to take place (1st) – manager can select to carry out the back up immediately... (1) – BECAUSE the back-up takes place at a time convenient to the manager/when the system is quiet (may explain why timing is convenient) (1) • Manager selects type of back-up (1st) – backing up only the specified data/files (1) – BECAUSE may want to back up whole device OR new files (1) • Manager selects how regularly the back-up is to take place (1st) – BECAUSE to minimise data loss (1) • Manager will decide where back-up is to be stored (1st) – BECAUSE to make sure back-up will be secure (1) • Manager selects whether to verify the back-up or not (1st) – BECAUSE to make sure the backup has been completed correctly (1) • Manager chooses whether or not to encrypt the back-up (1st) – BECAUSE adds an extra level of security (1) Must have at least TWO reasons to obtain full marks	5

Question	Answer	Marks
6(b)	MAX three from: • Can be either lossy or lossless. (1) • Lossless compression reduces number of bits (1st) – ...by identifying repeated patterns of data. (1) • No information is lost in <u>lossless</u> compression. (1) • <u>Lossy compression</u> reduces number of bits (1st) – ...by identifying unnecessary information. (1) Effects max three: • Quality of audio is the same when using <u>lossless compression</u> (1) • Quality of audio is lower when using <u>lossy compression</u> (1) • File size is smaller for lossy than lossless (accept converse) (1)	4

Mark scheme · 9626/11 May/June 2023 · Question 10

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Question	Answer	Marks
10	Five from: - Sampling resolution is the number of bits (1st) - per sound sample (1) - The higher the sampling resolution (1st) - the more accurately the (final) outcome (of the process) will represent the sound made (accept converse argument for whole answer) (1) - The higher the sampling resolution/the greater the resolution the greater the size of the file (accept converse) (1) - Modern day digital audio is normally found in 16 bit/24 bit resolutions 8-bit resolution can take one of 256 different values (1st) - which is not generally considered enough resolution to accurately represent music audio (1) 16-bit resolution can have one of 65 536 different values (1) 24-bit stores 16 777 216 different values (1) - the difference between 24 and 32- bit cannot be discerned by human hearing (1) - These values are all per channel (1)	5

Mark scheme · 9626/12 Oct/Nov 2023 · Question 10

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Question	Answer	Marks
10(a)	Two marks for an explained reason: e.g. - To remove unwanted/unnecessary parts at start/end (1) to make it less boring (1) - To shorten the length (1) because it is too long//to reduce file size (1) - To bring video and audio into synch (1) so that video looks more natural (1) - To make it more exciting (1) by removing boring introduction (1)	2

Question	Answer	Marks
10(b)	Two marks for an explained reason: e.g. • What has been said is shown on the screen//can be read (1) so that nothing is missed (1) • Viewers of a video may have hearing difficulties (1) so subtitles help them understand what is being said (1) • There may be a lot of background noise where the video is being watched (1) so that nothing is missed (1) • Video may be being viewed in a quiet area (1) where having the audio on would bother other people (1) • People watching the video might speak a different language//accent//be muttering (1) so subtitles in other languages can be provided (accept subtitles could be presented in the same language, so words can be read) (1)	2
10(c)	Two marks for an explained reason: e.g. • It can invoke an emotional response from the viewer (1) to increase engagement/enjoyment (1) • To grab audience attention (1) by making the video more interesting (1) • A narrator can provide further content (1) to drive home the emotion/mood//can clarify a video (1) • To add sound effect (1) can heighten the impact of (e.g.) explosions/speeding cars (1) to add authenticity/context/reality (1) • To add a clear soundtrack (1) without background noise (1)	2

Question	Answer	Marks
7(a)	Two from: - Fade in occurs at the beginning of a sound clip when the sound gradually gets louder (1) - Fade out occurs towards/at the end of a sound clip when the sound gradually gets softer (1) - Volume stops increasing when the required sound level is reached (1) - Sound disappears altogether (1)	2
7(b)	Two from: - Lossless compression retains the quality of the original/does not reduce the quality of the audio file (1) - Data is not lost (1) - When lossless compression software sees a repeated sequence of bits it replaces the repeated sequences with a character (1st) - which indicates what is being repeated and how many times (1)	2
7(c)	Two from: - Removes some of the audio file's data (1) - Reduces the quality of the audio file (1) - Removes sounds at very low and high frequencies (which are outside the range of human hearing) (1) - Creates smaller file sizes than lossless compression (1) - Once this data has been removed from the files it can never be replaced (1)	2

10(a)	Two available: - The number of samples (1st) - within a given period of time (accept ANY time unit) (1) - usually within 1 second (1).	2
10(b)	Max two for one complete explanation: - The sampling rate can affect the quality of audio files/how the audio sounds (1st) - because it captures sounds at discrete points (and not the whole sound) (1) - A higher sampling rate sounds better than a lower rate (1st) - because a higher sampling rate more accurately represents the original sound (1) - The lower the sample rate, the smaller the file size (1st) - because there is less data stored (1)	2

Mark scheme · 9626/12 May/June 2024 · Question 8

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Question	Answer	Marks
8(a)	Two from: - To remove unwanted parts of the clip//there are unwanted clips//parts (accept example) (1) - To reduce the length/time of the track//clip is too long (1) - To reduce file size//file size is too big//storage space required (1) - To make same length as an accompanying video clip//other example of fitting to an event (1)	2
8(b)	One from: - To make two (or more) audio clips flow into each other (1) - To combine the best parts of one clip with the best parts of another clip/combine the best parts of several recordings made from the same concert/act (1) - To reorder audio clips (1) - To give continuity/continuation//no gaps between them//sound files (1) - To create a single track/clip (1) - To make the file longer//a suitable length (1)	1
8(c)	Three from: - To boost/adjust the frequencies (in a sound signal) (1st) - By raising (1) or lowering (1) - Can use filters (1)	3

Mark scheme · 9626/13 May/June 2024 · Question 8

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Question	Answer	Marks
8(a)	Two marks available for a full explanation: To identify the people involved in the creation of the video (1) so they can get recognition//acknowledge their role in creating it (1)	2
8(b)	Two from: A frame from the video to be used as a photograph/thumbnail (1) used for (e.g.) marketing/publicity//to check for continuity//to be used as an icon (1)	2
8(c)	Two from: - Not all devices/platforms are able to play videos in all formats/the original format (1) - which means some videos are incompatible with specific devices/platforms - so this makes the video able to play/be accessed/used on a wider range of devices/platforms (1) - Videos may need to be converted to reduce their file size (1st) - to allow them to be attached to an email (1) - reduce storage space (1) - upload to video streaming website/social media (1)	2

Mark scheme · 9626/11 Oct/Nov 2024 · Question 11

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Question	Answer	Marks
11	FOUR from: - Identifies Lossless and identifies Lossy compression (1) Max 3 for Lossless: - Does not reduce the quality of the audio file (1) - Audio file can be recreated exactly as it was when first created (1) - Algorithms used to perform this compression mainly look for repeating patterns (1) Max 3 for Lossy: - Reduces quality of the audio file (1) - Creates smaller file sizes than lossless compression (1) - The data that is removed can never be replaced (1) - Some lossy compression software will remove frequencies that are outside the range of human hearing. (1)	4

Mark scheme · 9626/12 Feb/March 2025 · Question 10

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Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of two marks. - Sets the peak amplitude/loudest part of a track to a specific level. (1) - Makes multiple tracks have the same gain/peak/relative volume. (1) - Helps increase the volume of an entire audio track proportionally. (1) - Helps achieve consistent volume levels across different audio files. (1) - Helps maximising overall volume levels without compromising sound quality (1)	2
10(b)	One mark per bullet point to a maximum of two marks. - Eliminates/removes unwanted sounds. (1) - Makes audio track/signal clearer. (1) - Plays a mirror image of the original audio track (1st) - to partially cancel noise out. (1)	2

Question	Answer	Marks
6(a)	One mark per bullet point to a maximum of two marks. • To indicate time has passed between two scenes in a video (1) • To ease the emotions of the audience after a troubling scene (1) • To create suspense after/before a pivotal/important moment (1) • So that the audience are aware of the end/beginning. (1) • To provide a smooth transition between scenes (1)	2
6(b)	One mark per bullet point to a maximum of two marks. • To provide context • To provide information//better understanding (1) • Identifies author (1) • Identifies sound track (1) • Captions can help viewers engage with your video (1) • Search engines use metadata like captions to understand a video's content (1) • To provide translations (1) • To allow watchers to read what is being said//reason why what is being said may be inaccessible (1) Note This may be combined with “better understanding” to achieve two marks.	2

Question	Answer	Marks
8(a)	Max one for: - It means adding tracks/audio clips (1) Max one mark for any valid reason: e.g. - Add complexity to the music/audio (1) - Adds voiceover (1) - Add extra instruments (1) - Add sound effects to match what is happening on the screen (1)	2
8(b)	Max one for: - It means boosting/reducing/changing the levels of different frequency bands (1) Max one for any valid reason: e.g. - To pick out speech from background sounds (1) - To reduce the hiss (1) - To reduce/increase identified/intrusive noise frequencies (1) - To emphasise/highlight a guitar solo in a music track (1)	2
8(c)	Max one for: - Is an effect that adds echo to a sound (may describe sound being bounced off (e.g.) walls) (1) Max one for any valid reason: e.g. - To simulate being in a different environment (e.g. cavern/concert hall) (1) - To give depth/richness to a sound (1) - To create a dramatic effect in a radio/audio play/podcast (1)	2

Question	Answer	Marks
8(a)	One from: - Compressed data cannot be reconstructed (1) - It permanently removes data (1)	1
8(b)	One from: - Compressed data can be reconstructed (1) - Does not permanently remove data (1)	1

Question	Answer	Marks
8(c)	Four available. One per instance of a valid comparison: e.g. • Lossless will not affect the quality of video/lossy will affect the quality (1) • Lossy reduces the colour range (1) • Lossy can alter the frame rate (make the video jumpy/stutter) (1) • Lossy can desync the audio with video (1) • Lossy can affect the colour settings of the video (1) • Lossy can affect resolution of video (1) • Lossy can affect the bitrate/make incompatible with devices (1)	4

Question	Answer	Marks	Guidance
5(a)	ONE mark available: - It has a width of 16 units and height of 9 units (1) width to height ratio is 16:9/1.78:1 (1)	1	Do Not Award Length not allowed. Unless it is length and height
5(b)	TWO marks available for two complete explanations: - Ensures high-definition quality therefore final product will be of a high standard/clear to see (1) - Offers more content to the viewer so a better-quality product (1) - Is the more modern format, so product looks up-to-date (1) - Is the most widely used format in computer monitors/Television so the video will fit properly (1) 16:9 is the international/industry standard therefore it will be more widely used/meets international broadcast standards (1)	2	

Question	Answer	Marks	Guidance
6	TWO marks available: - Both use compression (1) - MP4a has a higher sound quality (1) - MP3 file sizes are larger (1) - MP4a has a higher bit rate (1) - MP3 is lossy/MP4a is lossless (1) MP3 files can be used on more platforms than MP4a (may exemplify a hardware type) OR MP3 Files are more widely used OR MP3 files is the standard format to use for sound files, MP4a is less widely used for sound files (1) Allow converse answers where applicable	2	